



PRL

A lifelong cognitive operating system — the second memory of a human being.

Human memory is limited, fragmented, and temporary.

PRL is a persistent, evolving memory system that grows alongside its owner — turning scattered life data into structured knowledge, and knowledge into wisdom.

Not a chatbot. Not a notes app. A second memory.

PRL continuously learns from a person's experiences, documents, habits, conversations, media, and actions — transforming scattered information into structured knowledge and long-term intelligence. Instead of merely *storing* information, it understands the relationships between memories, anticipates future needs, and preserves a person's knowledge for decades.

It exists to answer one fundamental problem:

"Human memory is limited, fragmented, and temporary. PRL creates a persistent and evolving memory system that grows alongside its owner throughout life."

What people lose every day

Documents

IDs, property papers, policies, certificates — misplaced.

Stories

Family history and wisdom, forgotten with a generation.

Records

Medical history and finances, scattered and fragmented.

PRL is a digital extension of human memory that never forgets and continuously organises life into meaning. The system transforms **data → understanding → wisdom**.

From data, to understanding, to wisdom.

Most tools stop at storage. PRL is built as a pipeline that keeps climbing the ladder of abstraction — each layer adds meaning the layer below could not see.

● Data

Raw signals: a commit, a bill, a prescription, a voice note, a photo, a chat.

● Information

Signals normalised into typed, time-stamped, embedded *memories*.

● Knowledge

Memories connected into a graph of people, projects, places & concepts.

● Wisdom

Patterns, predictions, blind-spots & a model that reasons the way you do.

A first principle: never fabricate

Every PRL insight is **evidence-grounded** — each one carries a score, a confidence, the evidence behind it, and a plain-language reason. When the data is too thin to support a claim, the system **says nothing** rather than inventing one. PRL measures; it never moralises.

One write path. One graph. Many minds.

Every signal — regardless of source — funnels through a single **Memory Engine**, gets embedded and entity-resolved, and is written once to a vector store and projected into a graph. Everything above reads from that shared substrate.

INGESTION · CONNECTORS

Capture reality from many sources

git free-text / journal filesystem browser shell documents* photos / audio* calendar*



MEMORY ENGINE · SINGLE WRITE PATH

Normalise → embed → resolve entities → persist

Pluggable embeddings (local sentence-transformers, 384-dim; offline hashing fallback). Every connector calls `ingest()`. Idempotent via dedupe keys.



STORAGE · MEMORY + KNOWLEDGE CORTEX

Vector memory (Postgres + pgvector) · Life Graph (Neo4j)

Semantic recall by meaning, and a relationship graph connecting people, projects, goals, skills & places — the brain's five regions.



COGNITION · THE REASONING SUITE

16+ evidence-grounded cognitive engines

Twin, prediction, trends, patterns, life-sim, identity, blind-spots, personal OS, knowledge-evolution, learning/decay, self-model, you-model.



LEDGER · LIFE-DOMAIN CORTEXES

Finance · Health · Family

Domain understanding mined from everyday text: bills & due-dates, medications & reminders, relationships & preserved stories.



INTERFACE

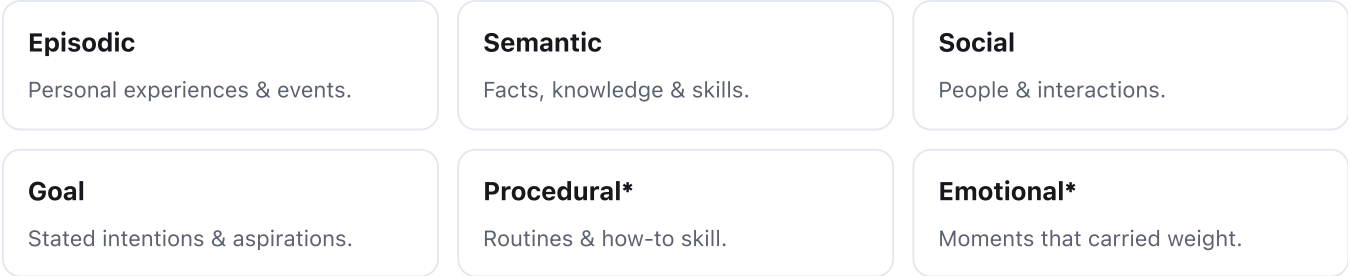
Cognitive Core (3D particle brain) · Self-Evolution Dashboard · Chat

Ask years of memory in natural language; watch the AI's internal state.

* Connectors marked with an asterisk are designed-for and on the roadmap; git, free-text, filesystem, browser and shell are implemented today.

Human-like memory, in layers.

Each memory is the atomic unit of reality: a time-stamped, embedded record with its source, importance, emotion, location and the entities it involves. Memories are typed into human-like layers so retrieval can reason about *what kind* of memory it is.



The Life Graph — five cognitive regions

Entities map onto the brain regions visualised in the Cognitive Core. Region intensity is data-driven — the more a region is fed, the brighter it burns.

● **Memory** (places) ● **Knowledge** (skills) ● **Goal** (goals) ● **Social** (people) ● **Project** (projects)

* Procedural and emotional layers are on the near-term roadmap; episodic, semantic, social and goal typing are implemented.

Sixteen ways of understanding a person.

Each engine is a pure, testable core plus a thin database adapter. All are explainable — score, confidence, evidence, reason — and all refuse to speak below their data threshold.

ENGINE	WHAT IT REVEALS
Cognitive Twin	Evidence-backed traits: learning style, persistence, curiosity, focus, attention rhythm.
Prediction	Probabilistic burnout, project completion, habit stability, knowledge decay.
Trends	Rising / fading / emerging / dormant interests over time.
Patterns	Cross-signal correlations (contemporaneous & time-lagged).
Life Simulation	30-day / 90-day / 1-year / 5-year trajectories + counterfactuals.
Life Chapters	Auto-detected phases from sustained shifts in dominant focus.
Identity	Claim vs. pursuit — stated priorities against observed behaviour.
Habit Genome	Each habit as a living entity: forming → growing → stable → declining → dormant → dead.
Decision Genome	Inferred decisions, their outcomes, and follow-through patterns.
Blind Spots	Hidden risks, patterns & opportunities synthesised across engines.
Personal OS	Current → desired → distance → obstacles → leverage points.
Knowledge Evolution	The temporal DAG of how concepts emerged and fed one another.
Learning / Decay	Per-concept retention curves, half-life, and at-risk knowledge.
Self-Model	What you say about yourself, reconciled against what behaviour shows.
You-Model	A from-scratch value model that weighs options the way <i>you</i> would.
Reflection / Insights	Plain-language life insights drawn from all of the above.

Finance. Health. Family.

Where the cognitive suite models *who you are*, the Ledger cortexes organise *what your life contains*. Each reads structured records mined from everyday text and returns an organised, evidence-backed view — and, like every PRL engine, stays silent when there is nothing to say.

● Finance Cortex

- Spend trend & top categories
- Recurring-bill detection
- **Next-due prediction**
- Savings rate
- Anomaly detection (robust median)

● Health Cortex

- Active medications
- Doctor visits
- Test-result trends
- Refill & check-up reminders
- record-keeping, not advice

● Family Cortex

- Relationship map
- Preserved stories & recipes
- Birthdays & anniversaries
- **Next-occurrence + reminders**
- Generational continuity

How a single sentence becomes structured memory

YOU WRITE	PRL EXTRACTS
"Paid ₹1,200 electricity bill"	expense · utilities · recurring → predicts next month's due date
"Dr. Sharma prescribed metformin; sugar 140"	prescription + test reading → med list, trend, check-up reminder
"Mom's birthday is on June 5"	family date → next occurrence + reminder; "Mom" → person in the graph

implemented All three cortexes, the text miners, the API endpoints (`/cortex/finance` · `/health` · `/family`), and 7 dedicated tests (suite: 29/29 passing) — verifiable offline.

From vault, to companion, to inheritance.

STAGE 1 · DIGITAL MEMORY

FOUNDATION BUILT

An intelligent vault

Store, search instantly, retrieve forgotten details, organise life automatically.

STAGE 2 · UNDERSTANDING

ENGINES LIVE

From remembering to recognising patterns

Spend trends, frequent contacts, health changes, work routines, productivity cycles.

STAGE 3 · PREDICTION

IN PROGRESS

From understanding to anticipation

Renewal reminders, bill due-dates, health-risk patterns, upcoming expenses, anniversaries.

STAGE 4 · COGNITIVE PARTNER

CHAT + YOU-MODEL

An intelligent companion that reasons over years

"Where's the 2017 property agreement?" · "When did I last see Dr. Sharma?" · "Which supplier is cheapest?"

STAGE 5 · LEGACY INTELLIGENCE

VISION

Memory becomes inheritance

Future generations interact with preserved stories, experiences and wisdom.

Your memory belongs to you.

Privacy-first architecture

- Local-first processing
- Encryption by default
- User-controlled permissions & selective sharing
- Transparent memory deletion
- No selling of personal data

PRL is not surveillance. It is ownership of personal intelligence.

Technology stack

Memory store	Postgres + pgvector
Life Graph	Neo4j
Embeddings	sentence-transformers (local)
Reasoning	from-scratch models + local LLM voice
API	FastAPI · Python
Interface	Next.js · React Three Fiber
Infra	Docker Compose · Redis

Roadmap

Multimodal intake

Documents (ID/property/policies), photos, audio, calendar.

Deeper memory

Procedural & emotional layers; richer retrieval.

Reality Interface

Timeline, life-graph & knowledge-galaxy front-end.

"To ensure that human experiences, knowledge, memories, and wisdom are never lost — but continuously evolve into intelligence that benefits both present and future generations."

PRL is not software. PRL is the digital continuity of human life.